Assistive Navigation and Obstacle Awareness for Visually Impaired Pedestrians: A Survey of Publications, Patents and Products, and Technology Trends

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Review 1 (DA1): domain introduction, problem statistics, comparative surveys of publications, IPRs and products, and technology trends.

Live demo video: https://drive.google.com/file/d/1yJS2f8_ee3ZwV55mkA9e1c6mm9f5xkYd/view?usp=sharing

***Abstract*—This review surveys assistive technology for pedestrians with visual impairment, the domain of a project that re-implements a recent cross-modal obstacle-avoidance wearable (WOAD) and extends it into a phone-based, offline, spoken guidance prototype (NeuroSonic). We substantiate the problem with WHO, Global Burden of Disease and Indian national survey statistics: at least 2.2 billion people have a vision impairment, 43.3 million were blind in 2020 and 61 million are projected by 2050, and 1.99 % of Indians aged 50 and over are blind. We compare published systems (wearable RGB-D and ToF devices, robot guides, segmentation methods), nine verified patents from 1981 to 2022, and eight commercial products from smart canes to free smartphone apps and robotic guides. The comparison shows that strong research results depend on custom hardware, that affordable products work on demand rather than continuously, and that continuous, motion-aware warnings from an ordinary phone are not described in the sampled literature, patents or products. We close with five technology trends (recognition, detection transformers, depth fusion, local offloading, and vision-language models with offline speech) and their adaptation status, which define the proposed work.**

***Index Terms*—assistive technology; visual impairment; electronic travel aid; obstacle detection; RF-DETR; multi-object tracking; camera-motion compensation; offline speech; pedestrian navigation**

I. Introduction: Domain and Field of Work

This work belongs to *assistive technology for people with visual impairment*, and within it to the sub-field of *electronic travel aids* (ETAs) and *navigation aids*. An ETA is a device that senses the environment ahead of a pedestrian and turns that information into a signal the person can hear or feel. A navigation aid tells the person where they are and which way to turn to reach a destination. The two needs are related but different: the first is about the next few metres and the next few seconds, the second about the whole journey [1], [2].

For decades the white cane and the guide dog have been the primary mobility aids. The cane is cheap, reliable, and gives immediate physical feedback about the ground, but its range is roughly the length of the cane, it does not detect obstacles at head or chest height, and it does not tell the user what an object is or whether it is moving. Guide dogs address some of these limits but are scarce and costly to train. The base paper of this project reports that, in its own trials, users with a white cane avoided only about a quarter of suddenly appearing indoor obstacles, and about 40 % of fast-moving cars at night [3]. These figures come from a specific experimental set-up, but they illustrate why supplementary sensing is studied at all.

The field draws on several engineering disciplines at once:

- **Sensing:** ultrasonic range finders, infrared and time-of-flight (ToF) depth cameras, ordinary RGB cameras, inertial sensors and satellite positioning (GNSS/GPS) [1].
- **Perception:** object detection, semantic and panoptic segmentation, tracking and scene description, today dominated by deep neural networks [4], [5], [6].
- **Human–computer interaction:** how to present information through hearing (speech, spatial audio, tones) or touch (vibration, braille) without overloading a sense that the user already relies on to move safely [1], [7].
- **Embedded and edge computing:** running perception within the power, weight, latency and cost limits of a wearable or a phone, sometimes with dedicated hardware such as FPGAs [3].
- **Navigation:** maps, routing and localisation, both outdoors (GNSS) and indoors (beacons, visual odometry) [8], [9].

Over the last decade, three changes have reshaped the field. First, deep-learning detectors made it possible to recognise many object classes in real time on modest hardware [4], [10], [11]. Second, smartphones became capable sensing and computing platforms that most users already carry, which lowers the cost and the social stigma of a dedicated device [2]. Third, large vision-language models have made free-form scene description available as a cloud service, which is now offered in consumer apps for blind users [12].

The present project is called **NeuroSonic**. It started from the base paper *WOAD: a wearable obstacle avoidance device for visually impaired individuals with cross-modal learning*, published in *Nature Communications* in 2025 [3]. WOAD combines self-developed glasses with a near-infrared camera and a ToF depth sensor, a reinforcement-learning video compression module, an FPGA, and a smartphone that runs a cross-modal (infrared + depth) obstacle detector. NeuroSonic re-implements and evaluates the WOAD detector, identifies the gaps that prevent it from being used with ordinary phones, and proposes and implements a phone-camera system with spoken guidance that runs offline on a laptop. This first review covers the domain, the size of the problem, and a comparative survey of publications, patents and products, followed by the technology trends that shape the proposed work.

The scope of this survey is deliberately bounded. It covers systems that sense the environment in front of a walking person and communicate with the person in real time, plus the navigation aids they are combined with. It does not cover

reading aids (text recognition for documents) except where they are part of the same product, and it does not cover medical treatment of vision loss. The patents and products listed are those that could be verified from primary pages (patent office records, publisher pages, manufacturer or established-retailer pages); they are a sample of the field, not an exhaustive census.

II. Problem Substantiation with Statistics

A. Global burden

The World Health Organization (WHO) estimates that, globally, at least **2.2 billion people** have a near or distance vision impairment, and that for at least **1 billion** of them the impairment could have been prevented or has yet to be addressed [13]. The 2.2-billion figure was first published in WHO's *World Report on Vision* in October 2019 [14]. Among people with distance vision impairment or blindness, the leading causes are cataract (94 million), refractive error (88.4 million), age-related macular degeneration (8 million), glaucoma (7.7 million) and diabetic retinopathy (3.9 million); near vision impairment is mainly due to presbyopia (826 million) [13]. Fig. 1 shows the distance-vision causes.

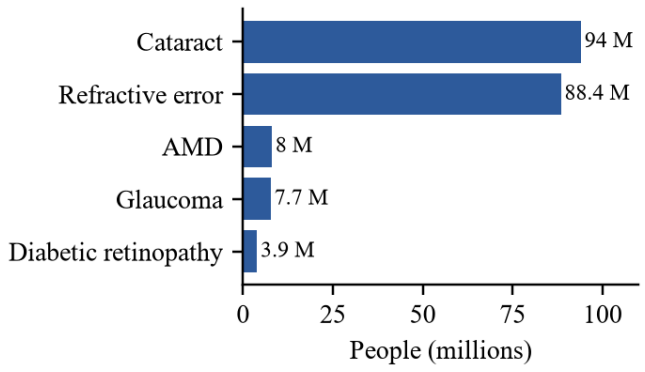


Fig. 1. Leading causes of distance vision impairment or blindness worldwide, in millions of people (WHO fact sheet, February 2026 [13]).

The Global Burden of Disease (GBD) Vision Loss Expert Group estimated that in 2020, **43.3 million** people were blind (95 % uncertainty interval 37.6–48.4 million), of whom 23.9 million (55 %) were female, and that 295 million people had moderate or severe vision impairment [15]. Between 1990 and 2020 the age-standardised prevalence of blindness among adults aged 50 years and older fell by 28.5 %, yet the *number* of blind people rose by 50.6 %, because populations grew and aged. The same study projects **61.0 million** blind people and 474 million with moderate or severe vision impairment by 2050 [15] (Fig. 2). In other words, medical progress is reducing the risk for each person, but the absolute number of people who need mobility support is still increasing.

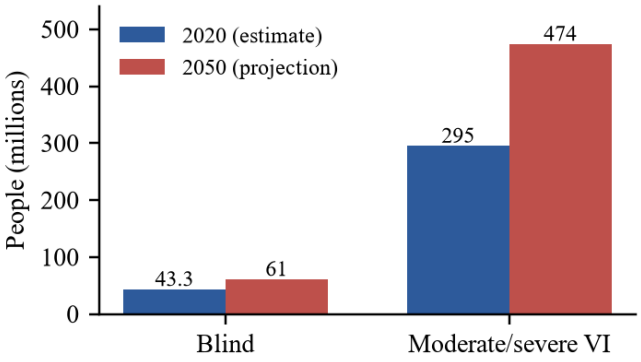


Fig. 2. People who are blind or have moderate/severe vision impairment (VI): 2020 estimate and 2050 projection, millions (GBD study [15]).

B. The situation in India

India's National Blindness and Visual Impairment Survey 2015–2019, run under the National Programme for Control of Blindness and Visual Impairment with AIIMS, surveyed people aged 50 years and older in 31 districts [16]. It reports a national prevalence of blindness of **1.99 %** in this age group, with cataract the major cause (66.2 % of blindness), and a 47.1 % reduction in blindness compared with the 2010 baseline [16], [17]. Prevalence rises steeply with age: 0.5 % at 50–59 years, 1.6 % at 60–69, 4.1 % at 70–79 and 11.6 % at 80 years and above [17] (Fig. 3). District values vary considerably, which matters for any local deployment [16].

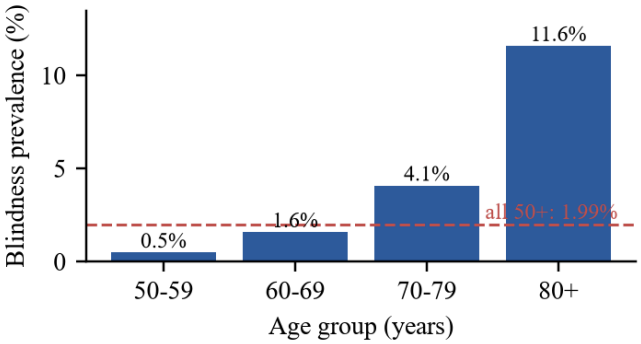


Fig. 3. Prevalence of blindness by age group among people aged 50+ in India (NBVIS 2015–2019 [17]). Dashed line: overall 50+ prevalence, 1.99 % [16].

C. What the statistics imply for this project

TABLE I
Key statistics and their relevance to the problem

Statistic	Value	Source	Implication
People with near or distance VI	≥ 2.2 billion	[13]	very large population; low-cost devices matter
Blind people (2020)	43.3 million	[15]	core users of mobility aids
Blind people projected (2050)	61.0 million	[15]	demand will grow
Blindness prevalence, India, age 50+	1.99 %	[16]	older users; simple interaction needed
Blindness prevalence, India, age 80+	11.6 %	[17]	age-friendly design (speech, few controls)
White-cane avoidance of sudden indoor obstacles (base-paper trial)	≈ 26 %	[3]	supplementary sensing can help

Three implications follow. *First*, the user population is large, growing and old. A device for this population must be affordable and must be operable by people who may not be

familiar with technology; speech is the most natural channel. *Second*, most of these users live in low- and middle-income countries, where the cost of dedicated hardware (several thousand US dollars for some commercial wearables, see the product survey) is a serious barrier, whereas a smartphone is often already owned. *Third*, because vision impairment is often partial and acquired late in life, people continue to rely on the cane. An electronic aid should supplement, not replace, the cane, and must never claim more than it can deliver. This last point is central to the design principles of NeuroSonic: it never states distances, never says that a way is "safe", and never claims 360-degree awareness.

The statistics do not, by themselves, measure the frequency of collisions or falls caused by obstacles while walking; we did not find a recent, population-level Indian statistic on that question that we could verify, so we do not quote one. The base paper's comparison with the white cane [3] is an experimental result from twelve participants, not a population statistic, and is quoted only as an illustration.

III. Survey: Comparative Analysis of Existing Solutions (Publications)

A. Method of the survey

Publications were selected to cover the main families of approaches: (i) review papers that map the field; (ii) wearable RGB-D or ToF systems with on-body or phone processing; (iii) robot guides; (iv) segmentation-based "walkable area" methods; and (v) the base paper. Each was checked against the publisher, PubMed Central or arXiv record. Where only the abstract was accessible, we report only what the abstract states and do not infer numbers. The comparison criteria are: *sensors*, *where the computation runs*, *perception method*, *user feedback*, *navigation support*, *evaluation*, and the *main limitation* relevant to a low-cost, phone-based system.

B. Review papers

Real and Araujo [1] review navigation systems for the blind from the first sonar-based ETAs and sensory substitution devices to smartphone, wearable and cloud-based designs. They identify recurring challenges: a heterogeneous user population, the limited bandwidth of hearing and touch, high cognitive load, poor acceptance of past devices, and a trade-off between positioning accuracy and infrastructure cost. Kuriakose, Shrestha and Sandnes [2] survey tools and technologies for navigation support and give design recommendations for future systems. Both reviews stress that technical performance alone does not guarantee adoption, which is why user-facing behaviour (what is said, when, and how reliably) is treated as a first-class requirement in this project.

C. Wearable depth-camera systems

WOAD (base paper) [3] is the most complete recent system. Self-developed glasses (about 400 g including an 80 g battery) carry a Synexens CS30 sensor with a near-infrared camera and a ToF depth sensor (640×480, 0.1–5 m). A depth-aided compression module, whose policy is learned with a discrete soft actor-critic reinforcement-learning agent [18], decides which frames to send and at what quality; it runs on a custom FPGA board. A smartphone (Samsung Galaxy S23 Ultra) runs a YOLOv5-based [19] detector extended with a second input stream and transformer blocks [20] that fuse infrared and depth features ("cross-modal learning"). Alerts are spoken (attention/warning/danger) and given by three vibration motors that encode direction and distance. The authors report 100 % collision avoidance in their indoor and outdoor scenarios, an end-to-end delay below 320 ms, about 11 h of operation, and a System Usability Scale score of 83.75 with twelve participants providing quantitative data [3]. It needs the custom glasses and the CS30 sensor.

Bai et al. [21] built a glasses-shaped travel aid with a consumer RGB-D camera and an inertial measurement unit. Ground segmentation separates the walkable ground from obstacles; a CNN on a smartphone classifies objects. The walkable direction is conveyed by beeps and object categories by speech. Twenty visually impaired participants navigated an office environment and avoided collisions; the abstract gives no quantitative accuracy or latency [21]. **Lin et al.** [22] combine an RGB-D wearable terminal, an earphone and a smartphone touch interface with a separate, powerful processor; a learned model predicts walkable instructions from RGB-D data and a semantic map. They report good results on indoor and outdoor datasets with low-lying obstacles and a user study, without numbers in the abstract.

Xu, Song and Wang [23] present a head-mounted obstacle-avoidance wearable (an extended version of a 2023 conference paper). **Tang et al.** [7] describe a human-centred wearable assistance system: an RGB-D camera on a 3D-printed glasses frame, ultrathin "artificial skins" with haptic actuators on the wrist and fingers, a bone-conduction earphone and smart insoles, with all computation local. It was tested with humanoid robots and with 20 visually impaired participants without prior experience of such systems; the authors argue that human adaptation has been under-considered in earlier systems [7]. **Yan et al.** [24] propose gait-based guidance using multimodal sensing and report better performance than a guide cane in indoor and outdoor experiments (preprint; no numbers in the abstract). **AviPer** [25] uses a visual-tactile multimodal attention network.

D. Robot guides and segmentation approaches

CaBot [26] is a suitcase-shaped autonomous navigation robot that guides blind users to destinations in unfamiliar indoor environments while avoiding obstacles, with a vibro-tactile handle; it was evaluated with ten blind users. It moves the sensing burden from the person to a robot, at the cost of carrying a robot. **Ghahremanians and Mahvash Mohammadi** [6] take a segmentation view: instead of detecting individual obstacles, a panoptic network finds the safe walking area and treats everything else as an obstacle. Their model reaches a panoptic quality of 37.6 on Cityscapes at about 4.3 frames/s (232 ms) on an RTX 4090, trained on a custom dataset that is not public.

TABLE II
COMPARATIVE ANALYSIS OF PUBLISHED SOLUTIONS

Work	Sensors	Compute	Perception	Feedback	Navigation	Evaluation (as reported)	Main limitation for a low-cost phone system
WOAD [3] (base)	NIR + ToF depth (CS30) on glasses	FPGA + smartphone	YOLOv5 + transformer cross-modal fusion	speech levels + 3 vibration motors	no route guidance	12 users; 100 % avoidance; <320 ms; ≈11 h	custom glasses and CS30 sensor required
Bai et al. [21]	RGB-D + IMU, glasses	smartphone	ground segmentation + CNN	beeps + speech	walkable direction	20 users, office	depth camera required
Lin et al. [22]	RGB-D wearable	separate processor + phone	learned walkable instructions, semantic map	earphone, touch	local instructions	datasets + user study	extra processor and depth camera
Xu et al. [23]	head-mounted sensors	wearable	obstacle detection	wearable alerts	—	see paper	dedicated device
Tang et al. [7]	RGB-D, artificial skins, insoles	local, on device	multimodal perception	bone conduction + haptics	guidance	robots + 20 users	several custom wearables
CaBot [26]	robot sensors	robot	autonomous navigation	vibro-tactile handle	indoor destinations	10 blind users	robot must be carried; indoor
Ghahremanians [6]	camera (smart glasses)	desktop GPU	panoptic segmentation of walkable area	—	—	PQ 37.6; 232 ms	no user study; dataset not public
NeuroSonic (this work)	phone RGB camera, mic, GPS	laptop GPU over Wi-Fi	RF-DETR + motion-compensated tracking + events	neural speech	offline map + A* routing	laptop tests only (no users yet)	needs laptop; no depth; not field-validated

E. Observations

- **Depth is the norm in research systems.** Nearly all wearable systems above use RGB-D or ToF sensors, which measure distance directly. A phone RGB camera does not, so a phone-based system must either avoid distance claims or estimate depth by other means.
- **Custom hardware limits reach.** Glasses, FPGAs, artificial skins and robots deliver strong results in studies but are not available to most users, especially in India.
- **Obstacle detection and navigation are rarely combined.** Most systems focus on the next few metres; route guidance is handled separately (e.g. by phone map apps), if at all.
- **Motion semantics are under-explored.** The surveyed detectors work frame by frame. Telling the user that a person is *moving into* the path, while not confusing this with the user's own walking, requires tracking and camera-motion compensation [27], [28].
- **Reproducibility is uneven.** Several works report results on private datasets; we found in re-implementing the base paper that its published dataset differs from the one behind its reported accuracy (Review 2).

IV. SURVEY: COMPARATIVE ANALYSIS OF IPRS (PATENTS)

A. Search method

Patents were searched on Google Patents with combinations of the terms *visually impaired*, *blind*, *obstacle detection*, *navigation*, *camera*, *audio* and *wearable*, and in the CPC group A61H 3/06 (walking aids for blind persons). For each candidate the bibliographic record was opened and the assignee, inventors, dates, legal status and summary were read. Nine documents were retained, chosen to span four decades and the main technical families. A search of the Indian Patent Office database was not completed in this review; for example, for IIT Delhi's SmartCane we could confirm the device and its funding but not a patent number, so it appears under products instead. A professional freedom-to-operate search would be needed before any filing (see the IDF in Lab 9).

B. The patents

US 4 280 204 (Polaroid, 1981) [29] describes a mobility cane with a capacitive electrostatic ultrasonic transducer that both emits and receives; one mode warns of low-hanging objects at upper-body height and another detects more distant objects. It is an early example of the ultrasonic ETA family, expired long ago.

US 2005/0208457 A1 (Fink and Humayun; later assigned to Caltech) [30] proposes a head- or eyeglass-mounted camera that captures an image on request, classifies the object with neural networks, look-up tables or thresholds, and speaks a description including colour and brightness. The application was abandoned. It anticipates, in a simple form, the camera-plus-speech idea of today's apps.

US 9 429 446 B1 (Searle et al., 2016) [31] claims a hand-held device that locates objects with a camera, radio, sonar or infrared module, plans a path and gives spoken directions. It lapsed in 2024 for non-payment of maintenance fees.

US 9 846 231 B2 (STMicroelectronics, 2017) [32] claims a wearable obstacle detector with an ultrasonic source and receiver plus a three-axis accelerometer; the control module enables the ultrasound and the time-of-flight measurement only when the foot rests on the ground during walking, then computes the obstacle distance and raises an alarm within a given range. The idea of gating sensing by gait phase is relevant to camera systems too, because image quality and apparent motion also depend on the walking cycle.

US 9 993 384 B1 (Toyota, 2018) [33] guides a user towards two destinations simultaneously with two distinct audio cues whose pitch or volume varies with the alignment between the user's heading and each path.

US 10 299 982 B2 (Frankel, 2019) [34] uses a smartphone or tablet camera and orientation sensors to build a 3-D model of a room; touching a spot on a top-down view announces the

object there and its attributes, such as potential hazards. Expired in 2023.

US 10 809 079 B2 (Ford, 2020) [35] describes a wearable garment such as a vest with GPS receivers, cameras, an ultrasonic sensor and a microphone that detect static and dynamic obstacles to generate and revise routes, with audio instructions and haptic cues, and optional upload of dynamic-obstacle data to update a map.

US 11 112 261 B2 (Brown University, 2021) [9] covers wearable navigation devices with multiple fixed cameras that localise the user by matching features against prior images and by visual-inertial odometry, and haptic belts or vests that render directional cues.

US 11 318 050 B2 (American Printing House for the Blind, 2022) [8] uses a smartphone's location services (GPS, compass, accelerometer, gyroscope) to narrate only the streets and points of interest in the direction the device is pointed, and Bluetooth beacons for indoor position and floor, with audio and haptic or braille output.

TABLE III
COMPARATIVE ANALYSIS OF PATENTS

Patent	Assignee	Year (grant/publ.)	Status*	Sensing	Processing / key idea	Output	Relation to NeuroSonic
US 4,280,204 [29]	Polaroid	1981	expired	ultrasonic transducer on cane	two range modes (upper body / distant)	warning signal	range sensing only; no recognition
US 2005/0208457 [30]	Caltech (assignee of record)	2005	abandoned	head-mounted camera	object classification on request	speech description	single images on demand; no tracking or path logic
US 9,429,446 [31]	individual	2016	lapsed 2024	camera/radio/sonar/IR	path planning	spoken directions	generic; no motion compensation
US 9,846,231 [32]	STMicroelectronics	2017	see record	ultrasound + accelerometer	sensing gated by gait phase; ToF distance	alarm by distance range	gating idea relevant; different sensor
US 9,993,384 [33]	Toyota	2018	active (per record)	heading sensors	two simultaneous audio cues	stereo audio	navigation audio design; no vision
US 10,299,982 [34]	individual	2019	expired 2023	phone camera + orientation	3-D room model, touch exploration	speech on touch	static room model, not live walking
US 10,809,079 [35]	Ford	2020	active (per record)	GPS, cameras, ultrasonic, mic on vest	static/dynamic obstacles for routing	audio + haptic	closest in scope; custom garment
US 11,112,261 [9]	Brown University	2021	active (per record)	multiple cameras + inertial	visual localisation	haptic belt/vest	localisation, not obstacle events
US 11,318,050 [8]	American Printing House for the Blind	2022	active (per record)	phone location sensors, beacons	direction-filtered narration of POIs	audio, haptic, braille	navigation narration; no camera

* Legal status as shown on Google Patents on 2026-11-03; Google notes this is an assumption, not a legal conclusion.

C. Observations from the patent survey

- **Two clusters.** Patents either sense obstacles (ultrasound, cameras) or support navigation (location narration, haptic direction), with the Ford vest [35] as the main example combining both, on a dedicated garment.
- **Several early patents have lapsed** [29], [31], [34], and the camera + speech concept itself appears in an abandoned 2005 application [30]; the general idea of "camera, recognise, speak" is therefore not novel and cannot be claimed.
- **What the sampled patents do not describe** is the specific combination pursued in NeuroSonic: an unmodified phone browser as a thin client; camera-motion-compensated tracking that announces only objects moving *relative to the background* into or across a walking-path region; suppression of announcements during strong camera motion; and offline speech, commands and routing on a nearby computer. Whether this combination is patentable can only be judged after a full search; this is the subject of the invention disclosure (Lab 9).

V. Survey: Comparative Analysis of Company Products

Products were chosen to represent the four commercial categories: smart canes, wearable cameras, smartphone apps and robotic or harness guides. Prices are quoted only where an established retailer or a consumer organisation published them, and they change frequently; they are indicative.

A. Smart canes

WeWALK is a smart handle that attaches to a standard white cane. It detects obstacles above chest level with an ultrasonic sensor and warns with vibration, connects to a smartphone, and has a touch pad for features such as public-transport information; battery life is quoted as up to five hours, and AARP lists it at US$599 [36], [37]. **SmartCane**, developed at IIT Delhi's Assistech laboratory with the Saksham Trust and manufactured by Phoenix Medical Systems, attaches to a white cane and uses ultrasonic sensing to detect obstacles above knee height, signalling distance with vibration patterns; it has long-range (about 3 m) and short-range (about 1.8 m) modes, and its translational development was funded by the Wellcome Trust [38]. Smart canes extend the cane's range upwards, but they measure distance only; they do not recognise objects or describe motion.

B. Wearable cameras

OrCam MyEye is a small camera that clips magnetically onto eyeglasses and works offline, without a smartphone or Wi-Fi; it reads text and, in the Pro version, recognises faces, colours and bank notes, controlled by gestures and voice commands. AARP lists the MyEye Pro at US$4,250; RNIB lists the Pro at £3,700 and the cheaper Smart version at £2,700 (both ex VAT) [36], [39]. It is a reading and recognition aid rather than a mobility aid.

C. Smartphone apps

Microsoft Seeing AI is free, with no subscriptions or in-app purchases, and organises tasks into "channels" (reading text, identifying products, recognising currency) [12]. **Google Lookout** (Android) offers several modes, including an Explore mode that identifies objects on the device, plus text and product recognition [40]. **Be My Eyes** connects a blind user by live video to a sighted volunteer, and its *Be My AI* feature describes a photo and lets the user ask questions about it [12]. These apps are free or low-cost and use a phone the user already has, but they work on single photos or on demand, need the user to point the phone, and several of their richer functions need the internet.

D. Harness and robotic guides

biped.ai NOA is a shoulder-worn harness with depth cameras covering a wide field of view (a 170° field is reported) that predicts object trajectories a few seconds ahead and warns with 3-D sound; a RealSense case study states that it uses the D430 depth module [41]. **Glidance Glide** is a two-wheeled, self-guided mobility aid pushed by the user, which uses cameras and sensors to steer around hazards; prices reported in early 2025 were US$1,500 plus a monthly subscription, and it was still in development and testing at the time of the reports [42].

TABLE IV
COMPARATIVE ANALYSIS OF COMPANY PRODUCTS (INDICATIVE; CHECKED 2026-11-03)

Product	Category	Sensing	Recognition	Motion / path events	Navigation	Offline	Indicative price
WeWALK [36], [37]	smart cane handle	ultrasonic	no	no	via phone app	obstacle sensing yes	US$599
SmartCane [38]	smart cane add-on	ultrasonic	no	no	no	yes	not verified
OrCam MyEye Pro [36], [39]	wearable camera	camera	text, faces, colours, notes	no	no	yes	US$4,250
Seeing AI [12]	phone app	phone camera	text, products, currency, scenes	no	no	partly	free
Google Lookout [40]	phone app (Android)	phone camera	objects, text, products	no	no	partly (Explore on device)	free
Be My Eyes / Be My AI [12]	phone app	phone camera	human volunteer; AI photo description	no	no	no	free
biped NOA [41]	shoulder harness	depth cameras	obstacles	trajectory prediction	GPS guidance (reported)	see vendor	not verified
Glidance Glide [42]	wheeled robot guide	cameras, sensors	hazards	steers around	yes	see vendor	US$1,500 + subscription (2025 reports)
NeuroSonic	phone + laptop prototype	phone camera, mic, GPS	23 obstacle classes	entering / crossing events	offline map, A* routing	yes (all local functions)	existing phone + laptop

E. Observations from the product survey

- **A price gap.** Dedicated wearables cost from about US$600 (cane handle) to over US$4,000 (wearable camera), whereas the phone apps are free. A system that uses hardware people already own addresses the affordability issue raised by the statistics.
- **A capability gap in the free apps.** The free apps recognise and describe, but on demand; they do not continuously warn about objects moving into the walking path. Continuous, motion-aware warning is offered only by dedicated hardware such as NOA.
- **Connectivity.** Richer descriptions depend on the internet; mobility support should not, because connectivity is unreliable exactly where people walk.

VI. TECHNOLOGY TREND: INTRODUCTION AND ADAPTATION STATUS

Fig. 4 places the surveyed documents on a timeline. Five technology trends run through them. For each we describe the technology, its state of adoption in assistive products, and how it is adapted in this project.

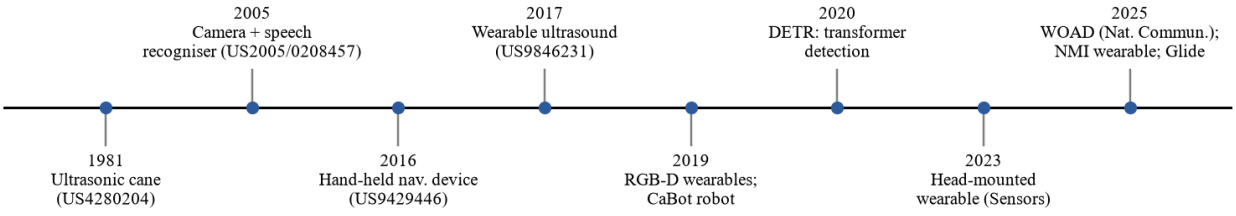


Fig. 4. Timeline of the surveyed patents, publications and products (positions evenly spaced; years from the cited documents): from ultrasonic canes to camera-based, learning-based wearables.

A. From range sensing to recognition

Early ETAs measured distance with ultrasound [29], [32], and ultrasonic smart canes are still the most widely available products [37], [38]. Since about 2016, deep convolutional detectors such as YOLO [4], SSD [10] and mobile networks such as MobileNetV3 [11] made it feasible to name objects in real time. Products like Seeing AI and Lookout show that recognition is now *commercially mature* for on-demand use [12], [40]. *Adaptation here:* NeuroSonic detects 23 obstacle-relevant classes from the COCO dataset [43] in every frame, rather than on demand.

B. Transformer-based detection

DETR [5] reformulated detection as set prediction with transformers [20], removing the hand-tuned non-maximum suppression step. Real-time variants now exist; RF-DETR [44] builds on the self-supervised DINOv2 backbone [45] and is released under the permissive Apache-2.0 licence. The base paper also uses transformer blocks, but for *fusion* of two modalities inside a YOLOv5 detector [3]. *Status:* research and open-source tools are mature; we did not find a surveyed assistive product that states it uses a detection transformer. *Adaptation here:* four detectors were benchmarked on the same images, and RF-DETR Nano was selected for its accuracy, recall and licence (Review 3).

C. Multi-modal and depth sensing

Research wearables overwhelmingly combine RGB or infrared with depth [3], [21], [22], [7], and the NOA harness brings depth cameras to a product [41]. Depth gives distance directly, which simplifies warnings. *Status:* established in research, early in products, and expensive. *Adaptation here:* the phone camera has no depth stream that a browser can use, so NeuroSonic deliberately makes no distance statements and uses image-space motion instead; monocular depth estimation is listed as future work.

D. Edge, offloading and dedicated hardware

Systems differ in where they compute: on an FPGA plus phone [3], on a separate processor [22], fully on the wearable [7], or in the cloud [12]. Offloading to a nearby, more powerful computer over a local network is a middle path: lower latency and better privacy than the cloud, more capacity than a phone. *Status:* standard in robotics and gaming, uncommon in assistive products. *Adaptation here:* the iPhone is a thin browser client; a laptop GPU on the same Wi-Fi performs detection, tracking, speech recognition, speech synthesis and routing, with no internet required.

E. Vision-language models and offline speech

Large multimodal models now describe whole scenes in natural language and answer questions about a photo, as in Be My AI [12]; APIs such as Gemini [46] make this available to developers. They depend on the internet and can state things with unjustified confidence. In parallel, offline speech technology has improved: Kaldi-based Vosk recognises speech locally [47], and Piper [48], built on VITS [49], synthesises natural speech on a CPU. *Status:* cloud description is commercially deployed; high-quality offline speech is mature in open source. *Adaptation here:* local speech in and out is the default; cloud scene description is optional, opt-in, rate limited and filtered so that it never states distance or safety.

TABLE V
TECHNOLOGY TRENDS AND ADAPTATION STATUS

Trend	Maturity in products	Adapted in NeuroSonic
Ultrasonic range sensing	mature (smart canes)	no (cane remains the user's primary aid)
CNN object recognition	mature (apps, on demand)	yes, continuous, 23 classes
Detection transformers	open source; not seen in surveyed products	yes (RF-DETR Nano, FP16)
Depth / multi-modal fusion	research; early products	no (base paper only); future work
Multi-object tracking	robotics; trajectory prediction in NOA	yes, with camera-motion compensation
Local offloading (LAN)	uncommon	yes (phone → laptop GPU)
Vision-language description	deployed (cloud)	optional, opt-in
Offline neural speech in/out	open source	yes (Piper, Vosk)
Offline maps and routing	common in navigation apps	yes (OpenStreetMap, A*)

VII. Synthesis and Conclusion of the Survey

The survey answers three questions that shape the rest of the project.

Is the problem significant? Yes. At least 2.2 billion people have a vision impairment, 43.3 million were blind in 2020 and 61 million are projected by 2050, and in India about 2 % of people aged 50 and over are blind, rising to 11.6 % above 80 [13], [15], [17]. The population is large, ageing and largely in settings where cost matters.

What do existing solutions leave open? Research systems achieve strong results with dedicated hardware: depth glasses, FPGAs, haptic skins and robots [3], [7], [26]. Products split into affordable but limited smart canes, expensive wearables, and free phone apps that work on demand rather than continuously [36], [12]. Patents cover range sensing, camera recognition with speech, and navigation narration, but the sampled patents do not describe continuous, camera-motion-compensated warnings of objects moving into the walking path from an unmodified phone [35], [30], [8].

Which technologies are ready to use? Real-time detection transformers with permissive licences, standard multi-object tracking, offline neural speech recognition and synthesis, and open map data are all mature [44], [28], [47], [48], [50]. Depth on ordinary phones and dependable cloud description are not yet ready for a mobility aid.

Direction for the project. Re-implement and verify the base paper (Review 2); then build a system that uses an ordinary phone as camera, microphone, GPS and speaker; runs perception, speech and routing offline on a nearby laptop GPU; warns about objects moving into or across the walking path while ignoring the user's own movement; and states honestly what it can and cannot see (Review 3). The system is a supplement to the white cane, not a safety device.

The main limitation of this survey is its breadth: it is a structured sample rather than a systematic review, the patent search was limited to US documents on Google Patents, and

product details come from manufacturer-adjacent or retailer sources that change over time. These limits are stated so that the comparisons are read as indicative.

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